

Krivine-Stengle Positivstellensatz analog for universal-quantifier-defined polynomial positivity sets

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Website #7 | Solver verdict: SOLVED | Verifier verdict: ISSUES | Shown: solver output

Problem

Let

$$K = \{x \in \mathbb{R}^n : g_j(x, y) \geq 0 \text{ for every } y \in Q, j = 1, \dots, s\},$$

where $Q \subseteq \mathbb{R}^m$ is closed, each $g_j \in \mathbb{R}[x, y]$, and ν is a finite nonnegative Borel measure with $\text{supp}(\nu) = Q$ satisfying the coordinatewise multivariate Carleman condition. Set $g_0 := 1$, and define

$$M := \text{QM}[g, \nu] = \left\{ \sum_{j=0}^s \int \tau_j(x, y) g_j(x, y) d\nu(y) : \tau_j \in \Sigma^2[x, y] \right\} \subseteq \mathbb{R}[x].$$

The task is to give a Krivine–Stengle-type certificate, directly in terms of g , the universal quantifier over Q , and ν , characterizing when a polynomial $f \in \mathbb{R}[x]$ is nonnegative on K , and in particular when $K = \emptyset$, without assuming archimedeanity.

Alternative readings considered

- The displayed Carleman sum in the prompt starts at $d = 0$, where the exponent $-1/(2d)$ is undefined. I interpret this in the standard way as the condition for $d \geq 1$.
- If $Q = \emptyset$, the support hypothesis forces a degenerate zero measure. The intended nontrivial regime is $Q \neq \emptyset$; this is the case I treat. (If $Q = \emptyset$, then $K = \mathbb{R}^n$ and the quantified part disappears.)
- “Analogue of Krivine–Stengle” can mean either an exact denominator/preordering certificate or any necessary-and-sufficient algebraic certificate. In the full non-semialgebraic/non-archimedean setting, I prove a sharp perturbative certificate; in the archimedean case I recover exact membership in the quantified quadratic module M .

Related literature found

- Hu–Klep–Nie (2025), *Positivstellensätze and Moment Problems with Universal Quantifiers* — SOLVES. This is the closest match to the exact quantified setting of the problem; I used it only as orientation, not as a proof ingredient.
- de Jeu (2003), *Determinate multidimensional measures, the extended Carleman theorem and quasi-analytic weights* — T00L. Gives the L^p -density of polynomials under the Carleman hypothesis; this is the analytic engine behind my separation lemma.
- Klep–Magron–Volčič (2025), *Sums of Squares Certificates for Polynomial Moment Inequalities* — T00L. Corollary 6.14 gives a perturbative Positivstellensatz for *arbitrary* constraint families $S \subseteq \mathbb{R}[x]$, even infinite ones.

- **Prestel (2005), *Representation of real commutative rings* and Kuhlmann (2010), lecture notes — TOOL/BACKGROUND.** These record the archimedean quadratic-module representation theorem (Jacobi / Stone–Krivine–Kadison–Dubois form) used in the bounded case.
- **Promising but not sufficient by itself: abstract preordering Positivstellensätze — BACKGROUND.** They give exact certificates on the *real spectrum* of an infinitely generated preordering, but in the present non-semialgebraic setting the pointwise set K need not coincide with that real-spectrum positivity domain. This is why the perturbative theorem is the right non-archimedean tool here.

Proof sketch

The proof has three steps.

First, I show that the quantified quadratic module $M = \text{QM}[g, \nu]$ has exactly the correct positivity set:

$$K = \{x \in \mathbb{R}^n : p(x) \geq 0 \ \forall p \in M\}.$$

The easy inclusion is $K \subseteq K_M$. For the reverse inclusion, if $\hat{x} \notin K$, then some quantified constraint fails at some $\hat{y} \in Q$: say $g_j(\hat{x}, \hat{y}) < 0$. I build a nonnegative bump ϕ near $\hat{y}$ and then approximate ϕ in $L^2(\nu)$ by a sum-of-squares polynomial $\sigma(y)$. The corresponding integrated polynomial

$$x \mapsto \int \sigma(y) g_j(x, y) d\nu(y)$$

lies in M and is negative at $\hat{x}$. The only analytic input is that SOS polynomials are dense in the cone of nonnegative $L^2(\nu)$ -functions; I derive that from de Jeu’s L^4 -density of ordinary polynomials by approximating $\sqrt{\phi}$ and squaring.

Second, once $K_M = K$ is known, the archimedean case becomes a direct application of the standard archimedean quadratic-module theorem: if M is archimedean and $f > 0$ on K , then $f \in M$. From this one immediately gets the weak nonnegativity certificate $f \geq 0 \iff f + \varepsilon \in M \ \forall \varepsilon > 0$, and the exact infeasibility certificate $K = \emptyset \iff -1 \in M$.

Third, for the general non-archimedean case I apply the perturbative theorem of Klep–Magron–Volčič not to the original quantified data, but to the already-constructed set of constraints $S := M \subseteq \mathbb{R}[x]$. Since M is itself a quadratic module and its positivity set is exactly K , their theorem yields

$$f \geq 0 \text{ on } K \iff (\forall \varepsilon > 0)(\exists r) \ f + \varepsilon \Omega_r \in M,$$

where

$$\Omega_r(x) = \sum_{i=1}^n \sum_{k=0}^r \frac{x_i^{2k}}{k!}.$$

This gives the desired non-archimedean quantified Positivstellensatz, and plugging in $f = -1$ gives the infeasibility certificate.

Main result

Theorem 1. *Let $Q \subseteq \mathbb{R}^m$ be closed, let $g_1, \dots, g_s \in \mathbb{R}[x, y]$, let ν be a finite nonnegative Borel measure with $\text{supp}(\nu) = Q$ satisfying the coordinatewise Carleman condition, and let*

$$M := \text{QM}[g, \nu] \subseteq \mathbb{R}[x].$$

For $r \geq 0$, define

$$\Omega_r(x) := \sum_{i=1}^n \sum_{k=0}^r \frac{x_i^{2k}}{k!}.$$

Then:

- (i) The positivity domain of M is exactly K :

$$K = \{x \in \mathbb{R}^n : p(x) \geq 0 \ \forall p \in M\}.$$

- (ii) If M is archimedean, then every polynomial $f \in \mathbb{R}[x]$ satisfying $f > 0$ on K belongs to M .

- (iii) If M is archimedean, then for every $f \in \mathbb{R}[x]$,

$$f \geq 0 \text{ on } K \iff (\forall \varepsilon > 0) f + \varepsilon \in M.$$

In particular,

$$K = \emptyset \iff -1 \in M.$$

- (iv) Without any archimedean assumption, for every $f \in \mathbb{R}[x]$,

$$f \geq 0 \text{ on } K \iff (\forall \varepsilon > 0)(\exists r \geq 0) f + \varepsilon \Omega_r \in M.$$

In particular,

$$K = \emptyset \iff (\forall \varepsilon > 0)(\exists r \geq 0) -1 + \varepsilon \Omega_r \in M.$$

Proof

First note that the Carleman hypothesis implies all even coordinate moments are finite; by generalized Hölder, every mixed moment of ν is finite as well. Hence every polynomial in y belongs to $L^p(\nu)$ for every finite p , and every integral appearing below is well-defined coefficientwise.

Also, M is a quadratic module in $\mathbb{R}[x]$: it contains 1 (take $j = 0$ and a suitable nonnegative constant SOS), it is closed under addition, and if $u \in \mathbb{R}[x]$ then

$$u(x)^2 \left(\sum_{j=0}^s \int \tau_j(x, y) g_j(x, y) d\nu(y) \right) = \sum_{j=0}^s \int (u(x)^2 \tau_j(x, y)) g_j(x, y) d\nu(y) \in M.$$

Lemma 1 (SOS density under Carleman). *If $\phi \in L^2(\nu)$ satisfies $\phi \geq 0$ almost everywhere, then for every $\eta > 0$ there exists $\sigma \in \Sigma^2[y]$ such that*

$$\|\phi - \sigma\|_{L^2(\nu)} < \eta.$$

Proof. Normalize ν to a probability measure $\bar{\nu} := \nu/\nu(\mathbb{R}^m)$. This does not change the relevant L^p -density statements, because the $L^p(\bar{\nu})$ - and $L^p(\nu)$ -norms differ only by the constant factor $\nu(\mathbb{R}^m)^{-1/p}$.

By de Jeu's extended Carleman theorem (Invoked result 1; hypothesis audit: after normalization, the standard-basis Carleman condition is exactly the stated hypothesis), ordinary real polynomials are dense in $L^4(\nu)$.

Set $h := \sqrt{\phi} \in L^4(\nu)$. Choose a real polynomial p such that

$$\|p - h\|_{L^4(\nu)} < \delta,$$

where $\delta > 0$ will be chosen momentarily. Let $\sigma := p^2 \in \Sigma^2[y]$. Then

$$\sigma - \phi = p^2 - h^2 = (p - h)(p + h).$$

By Hölder,

$$\|\sigma - \phi\|_{L^2(\nu)} \leq \|p - h\|_{L^4(\nu)} \|p + h\|_{L^4(\nu)}.$$

Also,

$$\|p + h\|_{L^4(\nu)} \leq \|p - h\|_{L^4(\nu)} + 2\|h\|_{L^4(\nu)} < \delta + 2\|h\|_{L^4(\nu)}.$$

Hence

$$\|\sigma - \phi\|_{L^2(\nu)} < \delta(\delta + 2\|h\|_{L^4(\nu)}).$$

Choosing δ small enough gives $\|\sigma - \phi\|_{L^2(\nu)} < \eta$. □

Lemma 2 (The quantified module has the correct positivity set). *If*

$$K_M := \{x \in \mathbb{R}^n : p(x) \geq 0 \ \forall p \in M\},$$

then $K_M = K$.

Proof. The inclusion $K \subseteq K_M$ is immediate. Indeed, let $x \in K$, and let

$$p(x) = \sum_{j=0}^s \int \tau_j(x, y) g_j(x, y) d\nu(y) \in M.$$

For $j = 0$, $g_0 = 1$. For $j \geq 1$, because $x \in K$ we have $g_j(x, y) \geq 0$ for every $y \in Q = \text{supp}(\nu)$. Since each τ_j is SOS, $\tau_j(x, y) \geq 0$ for all (x, y) . Therefore every integrand is nonnegative on $\text{supp}(\nu)$, so $p(x) \geq 0$. Thus $x \in K_M$.

For the reverse inclusion, let $\hat{x} \notin K$. Then there exist $j \in \{1, \dots, s\}$ and $\hat{y} \in Q$ such that

$$g_j(\hat{x}, \hat{y}) < 0.$$

Set

$$h(y) := g_j(\hat{x}, y) \in \mathbb{R}[y].$$

By continuity of h , there exist $\rho > 0$ and $\lambda > 0$ such that

$$h(y) \leq -2\lambda \quad \text{for all } y \in B_\rho(\hat{y}).$$

Because $\hat{y} \in \text{supp}(\nu)$, every open neighborhood of $\hat{y}$ has positive ν -measure, so $\nu(B_{\rho/4}(\hat{y})) > 0$.

Define the continuous nonnegative bump

$$\phi(y) := \max\left\{0, 1 - \frac{16\|y - \hat{y}\|^2}{\rho^2}\right\}.$$

Then ϕ is supported in $B_{\rho/4}(\hat{y}) \subseteq B_\rho(\hat{y})$, and $\phi > 0$ on a smaller ball around $\hat{y}$. Hence

$$I := \int \phi(y) h(y) d\nu(y) \leq -2\lambda \int \phi(y) d\nu(y) < 0.$$

Since h is a polynomial and all moments of ν are finite, $h \in L^2(\nu)$, and $\|h\|_{L^2(\nu)} > 0$ because $h \leq -2\lambda$ on a set of positive ν -measure.

By the previous lemma, there exists $\sigma \in \Sigma^2[y]$ such that

$$\|\sigma - \phi\|_{L^2(\nu)} < \frac{|I|}{2\|h\|_{L^2(\nu)}}.$$

Then, by Cauchy–Schwarz,

$$\left| \int (\sigma - \phi)h \, d\nu \right| \leq \|\sigma - \phi\|_{L^2(\nu)} \|h\|_{L^2(\nu)} < \frac{|I|}{2}.$$

Therefore

$$\int \sigma(y)h(y) \, d\nu(y) = I + \int (\sigma - \phi)h \, d\nu < I + \frac{|I|}{2} < 0.$$

Now define

$$p(x) := \int \sigma(y)g_j(x, y) \, d\nu(y).$$

Because $\sigma \in \Sigma^2[y] \subseteq \Sigma^2[x, y]$, we have $p \in M$. Evaluating at $x = \hat{x}$ gives

$$p(\hat{x}) = \int \sigma(y)h(y) \, d\nu(y) < 0.$$

Thus $\hat{x} \notin K_M$. Hence $K_M \subseteq K$, and so $K_M = K$. □

Proof of the theorem. Part (i) is exactly the preceding lemma.

For part (ii), first observe that if $-1 \in M$, then $M = \mathbb{R}[x]$. Indeed, for any $u \in \mathbb{R}[x]$, both $(u+1)^2$ and $-(u-1)^2 = (u-1)^2 \cdot (-1)$ belong to M , hence

$$u = \frac{(u+1)^2 - (u-1)^2}{4} = \frac{(u+1)^2 + (-(u-1)^2)}{4} \in M.$$

So in that case the conclusion is trivial.

Assume therefore that $-1 \notin M$. Then M is a proper archimedean quadratic module. By the archimedean quadratic-module representation theorem (Invoked result 2; hypothesis audit: $A = \mathbb{R}[x]$ is a commutative unital real algebra, M is an archimedean quadratic module by assumption, and characters of $\mathbb{R}[x]$ are exactly point evaluations $p \mapsto p(a)$, so the theorem's positivity set X_M identifies with K_M , which equals K by part (i)), every polynomial strictly positive on K belongs to M . This proves (ii).

For part (iii), if $f \geq 0$ on K , then $f + \varepsilon > 0$ on K for every $\varepsilon > 0$, so part (ii) gives $f + \varepsilon \in M$. Conversely, if $f + \varepsilon \in M$ for every $\varepsilon > 0$, then every $x \in K$ satisfies $f(x) + \varepsilon \geq 0$ for every $\varepsilon > 0$, hence $f(x) \geq 0$. Thus

$$f \geq 0 \text{ on } K \iff (\forall \varepsilon > 0) f + \varepsilon \in M.$$

Taking $f = -1$, we get

$$K = \emptyset \iff -1 \in M.$$

(The forward implication is immediate from part (ii), since $-1 > 0$ on the empty set; the reverse implication follows because every element of M is nonnegative on K .)

For part (iv), apply the arbitrary-constraint perturbative Positivstellensatz of Klep–Magron–Volčič (Invoked result 3; hypothesis audit: take $S := M \subseteq \mathbb{R}[x]$, which is allowed because their theorem permits arbitrary, even infinite, S ; by part (i), $K(S) = K_M = K$; and since M is already

a quadratic module, the quadratic module generated by S is just M itself). Their theorem yields, with $\Phi_r = \Omega_r$,

$$f \geq 0 \text{ on } K \iff (\forall \varepsilon > 0)(\exists r \geq 0) f + \varepsilon \Omega_r \in M.$$

This is exactly the claimed statement.

Finally, substituting $f = -1$ into part (iv) gives

$$K = \emptyset \iff (\forall \varepsilon > 0)(\exists r \geq 0) -1 + \varepsilon \Omega_r \in M.$$

□

Self red-team

- **Counterexample attempt against exact membership without archimedeanity.** Take $Q = \{y^0\}$ a singleton. Then, after rescaling by the mass of ν , M becomes the ordinary quadratic module generated by the finitely many polynomials $g_j(x, y^0)$. If there are no constraints, $M = \Sigma^2[x]$. The Motzkin polynomial is nonnegative on $\mathbb{R}^2$ but not SOS, so one cannot hope for a theorem of the form $f \geq 0 \Rightarrow f \in M$ in full generality. This supports the perturbative formulation in part (iv).
- **Counterexample attempt against the positivity-set lemma.** The delicate point is whether the integral cone M might be too small to detect a violated quantified constraint at a point $\hat{x} \notin K$. The proof addresses exactly this: it converts one violated pointwise constraint $g_j(\hat{x}, \hat{y}) < 0$ into a single integrated certificate by localizing near $\hat{y}$ and then approximating that localizer by an SOS polynomial in $L^2(\nu)$.
- **Three likely referee objections.**
 1. “*You used density of polynomials, but you need density of SOS.*” I handled this explicitly by approximating $\sqrt{\phi}$ in L^4 and squaring; this converts ordinary polynomial density into SOS density in L^2 .
 2. “*Why is $g_j(\hat{x}, \cdot) \in L^2(\nu)$?*” Because the Carleman hypothesis gives finite coordinate moments of all orders, and generalized Hölder gives finiteness of every mixed moment. Hence every polynomial in y lies in $L^2(\nu)$.
 3. “*When applying the perturbative theorem to $S = M$, why is $QM(S) = M$?*” Because M is already a quadratic module: it contains 1, is closed under addition, and is closed under multiplication by squares. Therefore the smallest quadratic module containing M is M itself.
- **Boundary cases checked.**
 1. If Q is compact, the Carleman condition is automatic, and the theorem specializes to a quantified compact Positivstellensatz.
 2. If Q is a singleton, the theorem reduces to the classical archimedean quadratic-module theorem in part (ii), and to the general perturbative positivity theorem in part (iv).
 3. If $K = \emptyset$, part (iii) gives the exact certificate $-1 \in M$ in the archimedean case, while part (iv) gives the correct perturbative certificate in full generality.

- **A route that looked stronger but fails in this generality.** The abstract Positivstellensatz for preorderings gives exact denominator certificates on the *real spectrum* of a preordering. I tried to apply it to the preordering generated by M . The obstruction is that, for infinitely generated/non-semialgebraic data, positivity on the point set K need not coincide with positivity on the entire relative real spectrum. So that route does not by itself solve the pointwise problem posed here.

Invoked results

1. **de Jeu's extended Carleman theorem (T00L).** *Statement used.* Let μ be a positive Borel probability measure on $\mathbb{R}^m$. If, for the standard basis $e_1, \dots, e_m$, each univariate moment sequence

$$s_j(n) := \int x_j^n d\mu(x)$$

satisfies Carleman's condition

$$\sum_{d \geq 1} s_j(2d)^{-1/(2d)} = \infty, \quad j = 1, \dots, m,$$

then polynomials are dense in $L^p(\mu)$ for every finite $p \geq 1$.

Citation. de Jeu, 2003, *Annals of Probability*, Theorem 2.3.

Hypothesis-match audit. I apply this after normalizing ν to a probability measure $\bar{\nu}$. The prompt assumes exactly the standard-basis coordinatewise Carleman condition. Therefore the theorem applies to $\bar{\nu}$, hence also to ν , and yields the L^4 -density of ordinary polynomials used in Lemma 1.

2. **Archimedean quadratic-module representation theorem (T00L).** *Statement used.* Let A be a commutative unital real algebra and $M \subseteq A$ an archimedean quadratic module. If $a \in A$ satisfies

$$\alpha(a) > 0 \quad \text{for every } \alpha \in X_M := \{\alpha \in \text{Hom}(A, \mathbb{R}) : \alpha(M) \subseteq \mathbb{R}_{\geq 0}\},$$

then $a \in M$. For $A = \mathbb{R}[x]$, this is equivalently the statement

$$f > 0 \text{ on } K_M \implies f \in M.$$

Citation. Standard form due to Jacobi / Stone–Krivine–Kadison–Dubois; see Prestel, 2005, *Expositiones Mathematicae*, and Kuhlmann, 2010, Corollary 2.3; also Marshall, 2008, Theorem 5.4.4.

Hypothesis-match audit. Here $A = \mathbb{R}[x]$, which is a commutative unital real algebra. The cone $M = \text{QM}[g, \nu]$ is a quadratic module by direct verification. In part (ii) we assume M archimedean. Every algebra homomorphism $\alpha : \mathbb{R}[x] \rightarrow \mathbb{R}$ is evaluation at the point $(\alpha(x_1), \dots, \alpha(x_n))$, so X_M identifies with K_M . By Lemma 2, $K_M = K$. Hence the theorem applies exactly as stated.

3. **Perturbative Positivstellensatz for arbitrary constraint families (T00L).** *Statement used.* For any subset $S \subseteq \mathbb{R}[x]$, let

$$K(S) := \{x \in \mathbb{R}^n : s(x) \geq 0 \ \forall s \in S\},$$

and let $QM(S)$ be the quadratic module generated by S . Define

$$\Phi_r(x) := \sum_{i=1}^n \sum_{k=0}^r \frac{x_i^{2k}}{k!}.$$

Then for every $f \in \mathbb{R}[x]$,

$$f \geq 0 \text{ on } K(S) \iff (\forall \varepsilon > 0)(\exists r) f + \varepsilon \Phi_r \in QM(S).$$

Citation. Klep–Magron–Volčič, 2025, *Foundations of Computational Mathematics*, Corollary 6.14.

Hypothesis-match audit. I apply this with $S := M \subseteq \mathbb{R}[x]$. Their theorem explicitly allows arbitrary, even infinite, S . By Lemma 2, $K(S) = K_M = K$. Since M is already a quadratic module, $QM(S) = M$. Their Φ_r is exactly my Ω_r . Therefore the theorem yields part (iv) verbatim.

Self-confidence

3 — I checked the internal derivations carefully, and the only imported steps are standard theorems whose hypotheses I audited explicitly; the remaining risk is in the exact formulation of those external tools rather than in the algebra carried out here.

Bibliography

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\begin{verbatim}
@article{HuKlepNie2025,
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  title = {Positivstellensatz and Moment Problems with Universal Quantifiers},
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  year = {2003},
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  year   = {2005},
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